

AS stoichiometry homework 1 · mole foundations

CIE 9701 AS Chemistry · 14 MCQ · 14 marks

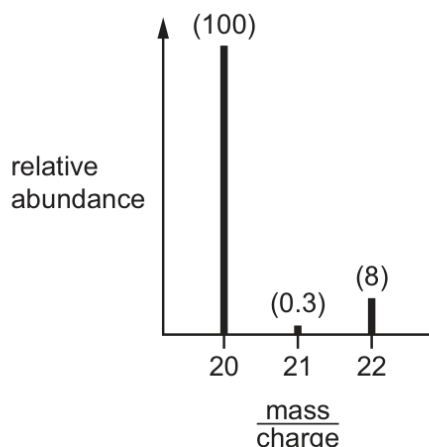
Name: _____ Class: _____ Date: _____

Topics: 1 masses of atoms and molecules · 2 the mole and Avogadro · 3 reacting masses · 4 complete combustion.

Circle **one** letter (A–D) for each question. Show working. Calculators and a Periodic Table may be used.

1 · Masses of atoms and molecules

1. The mass spectrum of a sample of neon is shown. The relative abundance of each peak is written in brackets above it.



What is the relative atomic mass, A_r , of this sample of neon?

- | | |
|----------------|----------------|
| A 20.15 | B 20.20 |
| C 21.00 | D 21.82 |

$$[1]$$

2. A sample of sulfur consists mostly of ^{32}S . It also contains 4.2% ^{34}S and 2.8% ^{36}S . No other isotopes of sulfur are present.

What is the relative atomic mass, A_r , of **this** sample of sulfur?

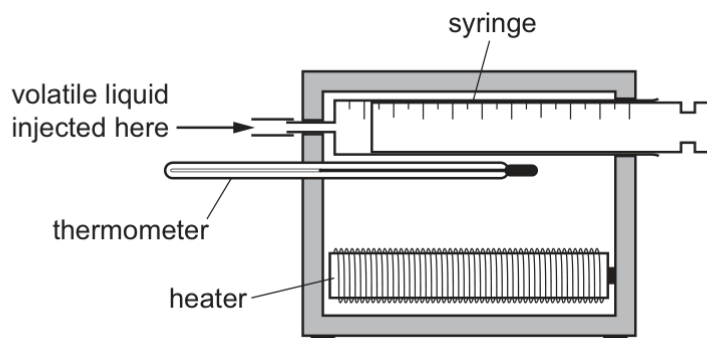
- A** 32.1 **B** 32.2
C 34.0 **D** 34.3

$$[1]$$

3. The diagram shows the apparatus used to find the relative molecular mass of a volatile liquid.

When 0.10 g of a volatile liquid is injected into the syringe, all of the volatile liquid evaporates and the volume increases by 85 cm^3 .

The heater maintains a temperature of 400 K and the experiment is carried out at a pressure of 101 300 Pa.



If the vapour of the volatile liquid behaves as an ideal gas, which expression can be used to calculate the relative molecular mass of the liquid?

- A.** $M_r = (85 \times 101\,300) \div (0.10 \times 8.31 \times 400)$
B. $M_r = (85 \times 101.3) \div (0.10 \times 8.31 \times 400)$
C. $M_r = (0.10 \times 8.31 \times 400) \div (85 \times 10^{-6} \times 101\,300)$
D. $M_r = (0.10 \times 8.31 \times 400) \div (85 \times 10^{-6} \times 101.3)$

$$[1]$$

2 · The mole and the Avogadro constant

4. Which statement about the Avogadro constant is correct?

- A.** It is the mass of one mole of any element.
B. It is the mass of 6.02×10^{23} atoms of any element.
C. It is the number of atoms in one mole of neon.
D. It is the number of atoms in 12 g of any element.

$$[1]$$

5. Which contains the largest number of hydrogen atoms?

- A.** 0.10 mol of pentane
B. 0.20 mol of but-2-ene
C. 1.00 mol of hydrogen molecules
D. 6.02×10^{23} hydrogen atoms

$$[1]$$

6. Which sample contains the most iodine?

- A** 1 g of CaI_2
- B** 1 g of KI
- C** 1 g of NaI
- D** 1 g of NH_4I

$$[1]$$

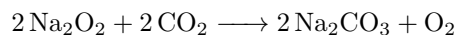
7. What contains 9.03×10^{23} oxygen atoms?

- A. 0.25 mol aluminium oxide
B. 0.75 mol sulfur dioxide
C. 1.5 mol sulfur trioxide
D. 3.0 mol water

$$[1]$$

3 · Reacting masses

8. Sodium peroxide, Na_2O_2 , is used to absorb carbon dioxide from the atmosphere and release oxygen in closed environments such as space capsules and submarines.



Which mass of sodium peroxide would be required to remove 2.4 dm^3 of carbon dioxide from the atmosphere at room temperature and pressure?

- A** 2.4 g **B** 3.9 g
C 7.8 g **D** 15.6 g

$$[1]$$

9. A student reacts 0.100 mol of each of sodium, magnesium and phosphorus atoms separately with an excess of oxygen.

Which rows are correct?

	oxide	mass of oxide formed / g
1	sodium	3.10
2	magnesium	4.03
3	phosphorus	7.10

- | | |
|-----------------------|-----------------------|
| A 1, 2 and 3 | B 1 and 2 only |
| C 1 and 3 only | D 2 and 3 only |

$$[1]$$

10. Separate samples, each of mass 1.0 g, of the compounds listed are treated with an excess of dilute acid. Which compound releases the largest amount of CO_2 ?

- A** 1.0 g CaCO_3
- B** 1.0 g Li_2CO_3
- C** 1.0 g MgCO_3
- D** 1.0 g Na_2CO_3

$$[1]$$

11. 2.0 g of ammonium nitrate, NH_4NO_3 , decomposes to give 0.90 g of water and a single gas.

What is the identity of the gas?

- A** NO **B** NO₂
C N₂O **D** N₂

$$[1]$$

4 · Complete combustion

12. What is the minimum mass of oxygen required to ensure the complete combustion of 12 dm^3 of propane measured under room conditions?

A 60 g

B 80 g

C 120 g

D 160 g

[1]

13. Mixture R consists of one mole of C_3H_6 and one mole of C_4H_6 .

What is the minimum number of moles of oxygen molecules needed for complete combustion of mixture R?

A 6.5

B 7

C 10

D 20

[1]

14. If 1 mole of hexane combusts in an excess of oxygen, how many moles of products are formed?

A 11

B 12

C 13

D 14

[1]